

I. Single choice (20%)

- The interrupt request () cannot be masked.
A. NMI B. READY C. INTR D. $\overline{MN}/\overline{MX}$
- In an 8086 system, 0F4H, 80H, 0B0H and 98H are stored in row from address 0000H:0040H. Which of the following statements is correct? ()
A. After execute 'INT 10H', IP= 80F4H B. After execute 'INT 10H', IP= 98B0H
C. After execute 'INT 40H', IP= 0F480H D. After execute 'INT 40H', IP= 98B0H
- If the decimal number -25 is encoded as 99H, the code is the () of -25.
A. sign and magnitude B. 1's complement C. 2' complement D. BCD code
- When 8086 is executing instruction 'MOV AL, [1001]', the control signals $\overline{RD}$, $M/\overline{IO}$, $\overline{BHE}$ are (), respectively.
A. 1, 0, 0 B. 1, 0, 1 C. 0, 1, 0 D. 0, 1, 1
- Which of the following statements about SRAM and DRAM is not correct? ()
A. DRAM is faster than SRAM. B. SRAM consumes more power than DRAM.
C. The memory density of DRAM is higher than SRAM D. DRAM needs to be refreshed regularly.
- 'LEA BX, VAR1' is equivalent to ().
A. MOV BX, VAR1 B. MOV BX, OFFSET VAR1
C. MOV BX, SEGMENT VAR1 D. MOV VAR1, BX
- If 4K×4bits SRAM chips are used to form a 32K×8bits SRAM memory, then the number of SRAM chips and the number of address lines required for this memory are (), respectively.
A. 16, 3 B. 8, 3 C. 16, 15 D. 8, 12
- Data transmission between an 8086 CPU and a UART 8251A is ____ communication and data transmission between a UART 8251A and another UART 8251A is ____ communication. ()
A. serial, serial B. parallel, parallel C. serial, parallel D. parallel, serial
- The instruction 'LOOP' of 8086 uses register () as a counter.
A. DX B. BX C. CX D. AX
- Once the 8086 CPU is reset, CS:IP = ().
A. 0000H:0000H B. 0FFFFH:0FFFFH C. 0FFFFH:0000H D. 0FFFFH:0FFFEH
- The immediate number 0100H in the "MOV AX, 0100H" instruction actually locates in the ().
A. data segment B. code segment C. stack segment D. extend segment

12. After the execution of an IRET instruction, the value of SP will increase by ().
- A. 2 B. 8 C. 6 D. 4
13. About interrupts in 8086 systems, which of the following statements is correct? ()
- A. Before entering an ISR, the CPU is set to disable all maskable interrupt requests.
B. In an ISR, CPU cannot execute an "INT" instruction.
C. If an 8259 is used to manage interrupt requests, an interrupt request with lower priority can preempt the current ISR as long as an EOI command is executed.
D. NMI can only be triggered by hardware.
14. Which of the following logical addresses cannot result in the physical address of 28EFFH? ()
- A. 28EFH:000FH B. 18F0H:0FFFFH C. 19FFH:0EF0FH D. 18FFH:0FE0FH
15. As to the memory and I/O address spaces in 8086 systems, which of the following statements is correct? ()
- A. We need 16-bit addresses to access both memory and I/O devices.
B. The I/O space is only 64K which is separate from the 1M memory space.
C. The I/O space is only a segment of the memory space.
D. DX has to be used in I/O instructions.
16. Which of the following statements is correct? ()
- A. The word size of a memory module depends on its addressable unit.
B. SRAM uses no capacitors to store data and therefore no refreshing circuit is required, which makes it suitable for permanent storage.
C. Though CISC instructions have variable length and execution time, the pipeline technique can still be used to improve the CPU performance.
D. When using DMA for data transfer between memory and some I/O device, the CPU will not be disturbed.
17. About the minimum mode of an 8086 CPU, which of the following statements is correct? ()
- A. All control signals are issued from the CPU.
B. The work mode of the CPU can be changed by a program.
C. Data transceivers such as 8286 are not required.
D. Bus control chip such as 8288 are required.
18. Which of the following statements about the concept of memory segment in an 8086 system is correct? ()
- A. There can be only one code segment but multiple data segments in a program.
B. A segment always starts on an address evenly divisible by 4.
C. 8086 can access at most 4 segments in a program.
D. A memory unit can be accessed from different segments.

19. If in the definition of a data segment ORG 200H comes before the definition of a block of data of 4KB and DS = 1002H, then the address of the last byte in this block is ().
- A. 11220H B. 1002H:0FFFH C. 1102H:01FFH D. 1022H:1000H
20. When using DMA, the data path between the main memory and a peripherals is ().
- A. memory $\leftrightarrow$ CPU $\leftrightarrow$ peripheral B. memory $\leftrightarrow$ CPU $\leftrightarrow$ DMA controller $\leftrightarrow$ peripheral
C. memory $\leftrightarrow$ DMA controller $\leftrightarrow$ peripheral D. memory $\leftrightarrow$ peripheral

II. Fill Blanks (20%)

- The range of an 16-bit binary number in 2's complement representation is from _____ to _____; Number 8000H in 2's complement representation represents an integer of _____.
- In the memory of an 8086 system, if 20H, 0EFH, 00H and 0F0H are stored from the address 0000H: 0100H, the corresponding interrupt number, CS and IP are _____, _____, and _____, respectively.
- In a computer system, if the width of the data bus and that of the address bus are 32-bit and 20-bit, respectively, and the addressable unit is also 32-bit, then the word size of this computer is _____ and the maximum capacity of its memory is _____ bytes.
- In a bus cycle of 8086/8088, there are at least _____ T states; during T1 and T2, AD[0:15] pins are used to convey _____ information and _____ information, respectively; Tw can be inserted if _____. For example, when 8086 executes the instruction: "MOV BYTE PTR [27H], 45H", M/ $\overline{IO}$ signal will change to _____ level; in the T1 period, data transferred on AD[0:7] are _____ H; in the T2 period, data transferred on AD[0:7] are _____ H. (Note that assume DS=200H)
- The Von Neumann architecture consists of five components, which are _____, _____, _____, input devices, and output devices, respectively.
- Read a piece of code as follows:

```
.data
ORG 10H
X DW 1020H
ORG 11H
Y DB '0304'
```

The contents stored at [0010H] and [0011H] are _____, _____, respectively.

III. Answer Questions (22%)

- Please describe how 8086 executes MOV [1001H], AX while receiving an INTR signal of INT type 40, given that CS=2000H, IP=0010H, DS=3000H, ES=4000H, AX=2327H, IF=1, and the four bytes stored from 0000H:00A0H are 20H, 30H, 00H, 21H, respectively. (14%)

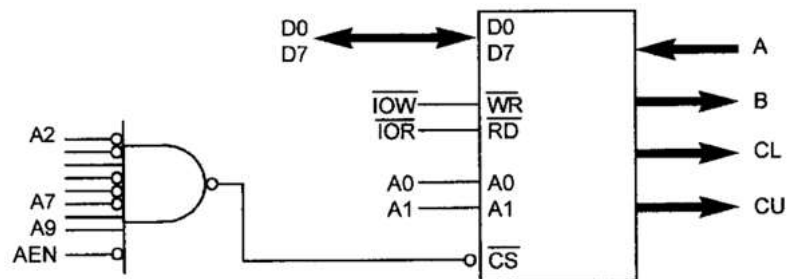
a) When will the CPU know there is an INTR? Will the CPU respond to this INTR and why? (3%)

b) How many bus cycles will the MOV instruction take? Please describe what the CPU does in each bus cycle. (4%)

c) After the execution of the MOV instruction, what are the values stored at 3000H:1001H and 3000H:1002H and what are the values of CS and IP? (4%)

d) Please tell the difference between the concept of system clock, bus cycle and instruction cycle using the above MOV instruction as an example. (3%)

2. If an 8255 is connected to a computer system as shown in the following figure, please tell whether the computer system is 8086 based or 8088 based and why? (4%)

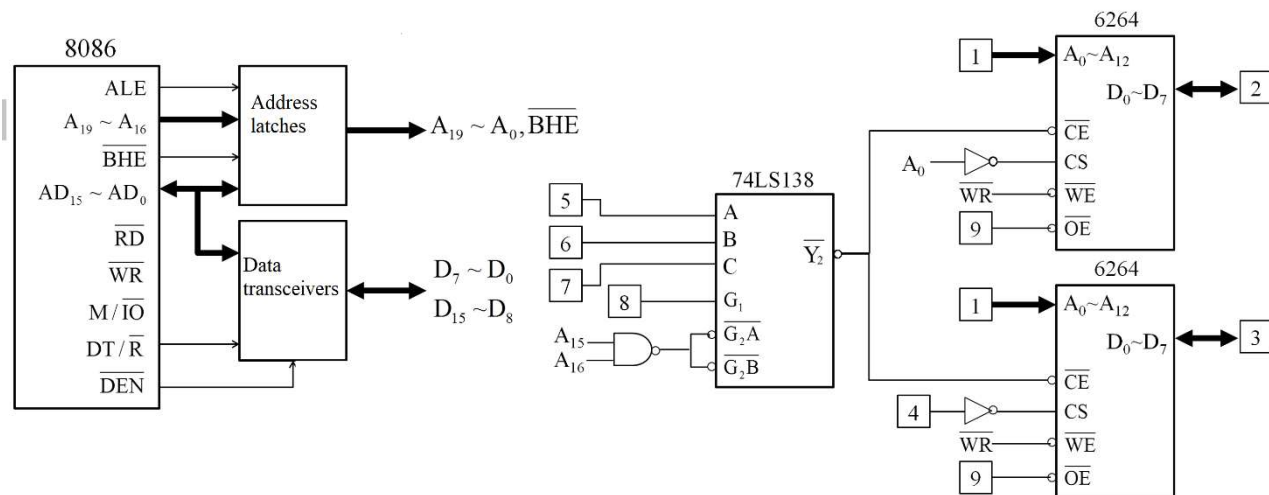


3. Convert the following C program into 8086 assembly codes. (4%)

```
int array[5];
int n = 5;
int sum = 0;
for (int i = 0; i < n; i++) {
    sum += array[i];
}
```

IV. Experiments (38%)

1. Examine the schematic below. An 8086 minimum system uses two chips of 6264 (8K bytes) to extend a 16K bytes 16-bit RAM memory. If the RAM chips are mapped into the address range B8000H~BBFFFH, try to fill the blanks 1-9 with proper CPU pin or output of address latch or output of bus drive. (9%)



1. _____	2. _____	3. _____
4. _____	5. _____	6. _____
7. _____	8. _____	9. _____

2. The following code is an initialization of 8253 (assume the port number of Counter 0, Counter 1, Counter 2 and the control register is defined as P8253_T0, P8253_T1, P8253_T2, and P8253_CS, respectively; the frequency of CLK₀ is 1MHz and OUT₀ is connected to CLK₁), which sets up Counter 0 to generate square wave with a frequency of 500Hz and Counter 1 to generate square wave with a period of 0.2 seconds. Please fill in all blanks. (10%)

```

INIT8253 PROC
    MOV AL, _____ B    ; 设定 Timer0
    MOV DX, _____
    OUT DX, AL

    MOV AX, _____
    MOV DX, P8253_T0
    OUT DX, AL
    MOV AL, AH
    OUT DX, AL

    MOV AL, _____ B    ; 设定 Timer1
    MOV DX, P8253_CS
    OUT DX, AL
    MOV AX, _____

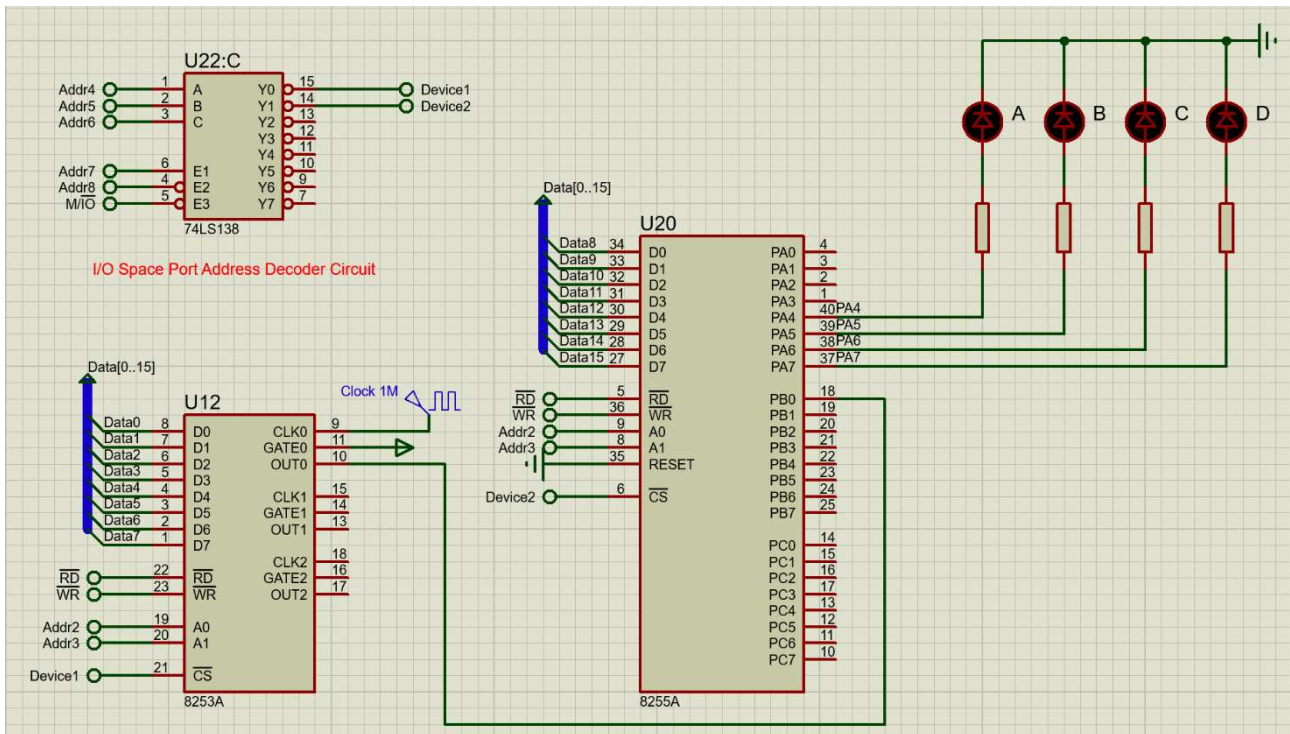
```

```

MOV DX, P8253_T1
OUT DX, AL
RET
INIT8253 ENDP

```

3. Examine the schematic below. Four LEDs are labeled A, B, C and D. The LEDs are switched on according to the position of current state within the state loop: $A \rightarrow B \rightarrow C \rightarrow D \rightarrow C \rightarrow B \rightarrow A \rightarrow \dots$. Each state stays 1ms and then steps forward within the loop. Please fill in the blanks below to implement the given function. (19%)



a) Initializing 8253: (6%)

```

MOV AL, _____
OUT _____, AL
MOV AL, _____
OUT _____, AL
MOV AL, _____
OUT _____, AL

```

b) Initializing 8255: (2%)

```

MOV AL, _____
OUT _____, AL

```

c) Polling the positive transition of PB0: (5%)

```

IN AL, _____
L0: CMP AL, _____
J____ L0
L1: CMP AL, _____
J____ L1

```

d) Write a program to step forward the state of the LEDs. (6%)